Business Intelligence

From:

- Taniar, D & Rahayu, W 2021, Data Warehousing and Analytics: Fueling the Data Engine. Data-Centric Systems and Applications, 1st edn, Springer, Cham Switzerland. https://doi.org/10.1007/978-3-030-81979-8
- Lay-Ki, S 2022, FIT3003 – Business Intelligence and Data Warehousing, Monash University, Australia

Business Intelligence Systems

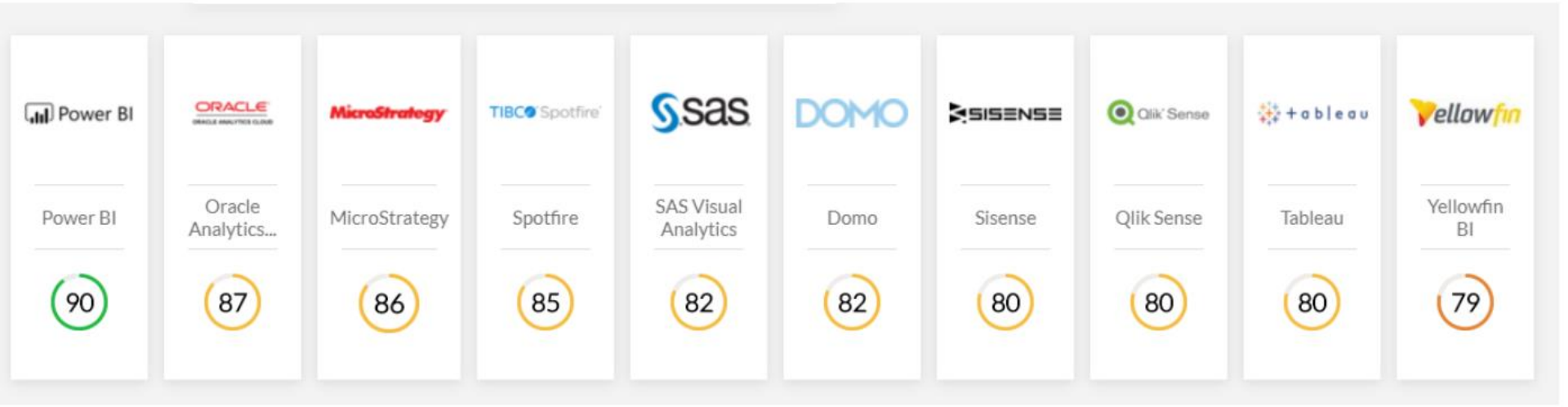
- What are business intelligence (BI) systems?
  - "*An integrated set of tools, technologies and programmed products that are used to collect, integrate, analyse and make data available.*" (Reinschmidt and Francoise, 2000)
- Use of technology for **decision support** not new:



- Functionally, EIS and BI very similar.
  - Current trend towards the use of the term 'Business Analytics'

BI Systems in the Market

Top 10 Business Intelligence Tools Leaders by Analyst Rating (of 269 products)



Source: selecthub.com

Business Intelligence Systems

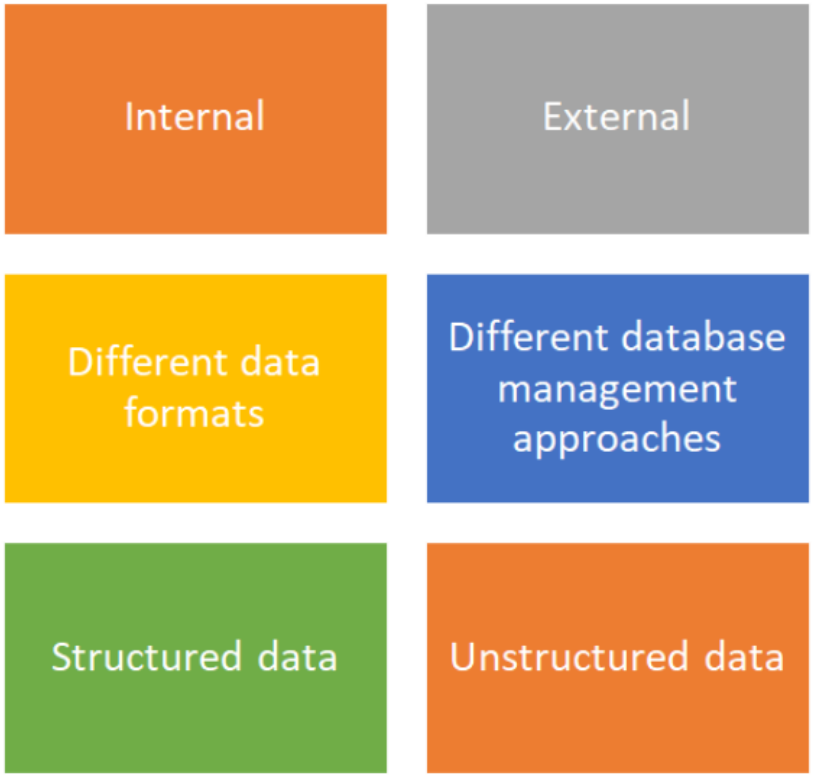
- Common features
  - Deals with huge amount of data, from different sources
  - Include dashboards that provide overview and monitoring of performance
  - Supports navigation from one report to the another for further investigation
  - Visualise data in both textual and graphical

Data and Business Intelligence

- BI systems have to:
  a. Acquire data
  b. Process it
  c. Present it

- Where does the data come from?

- Solution: data warehouse



Why learn about BI/DW?

- Three key aspects of BI/DW systems make them different to other information systems:
  a. The task supported
  b. The users
  c. The development process required

The Task - Decision Making

- Subject to
  - cognitive overload
  - systemic biases
  - personal preference

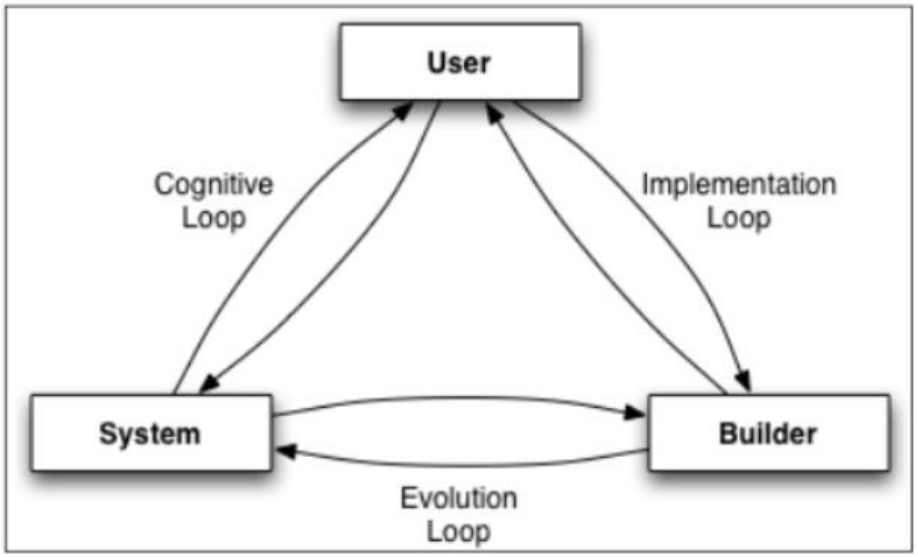
- The same decision task leads to different outcomes for different people!
  - Inherently ambiguous, especially for strategic decisions
  - No single ‘right’ answer
  - Possibly no single ‘right’ way to make a decision

The User

- Organisational decision makers:
  - Knowledge workers
  - Managers
  - Executives
  - Directors
- Impossible to specify what these people do from moment to moment
  - Compare with a sales clerk: very specific actions, repetitive task
  - Managers: many different actions, sometimes unique one-off tasks
- Organisationally powerful

The Development Process (cont.)

Framework that supports any system that supports decision making:



Peter Keen, 1980

User Interface in BI Systems

- The user interface must be easy to learn and easy to remember
  - Managers will not attend long training sessions
  - Some managers only use the system infrequently

- Interface must not get in the way
  - Allow analysis “at the speed of thought”
  - “Transparent interaction”

- Information presented must be easy to understand
- It must be clear where information comes from

Dashboards

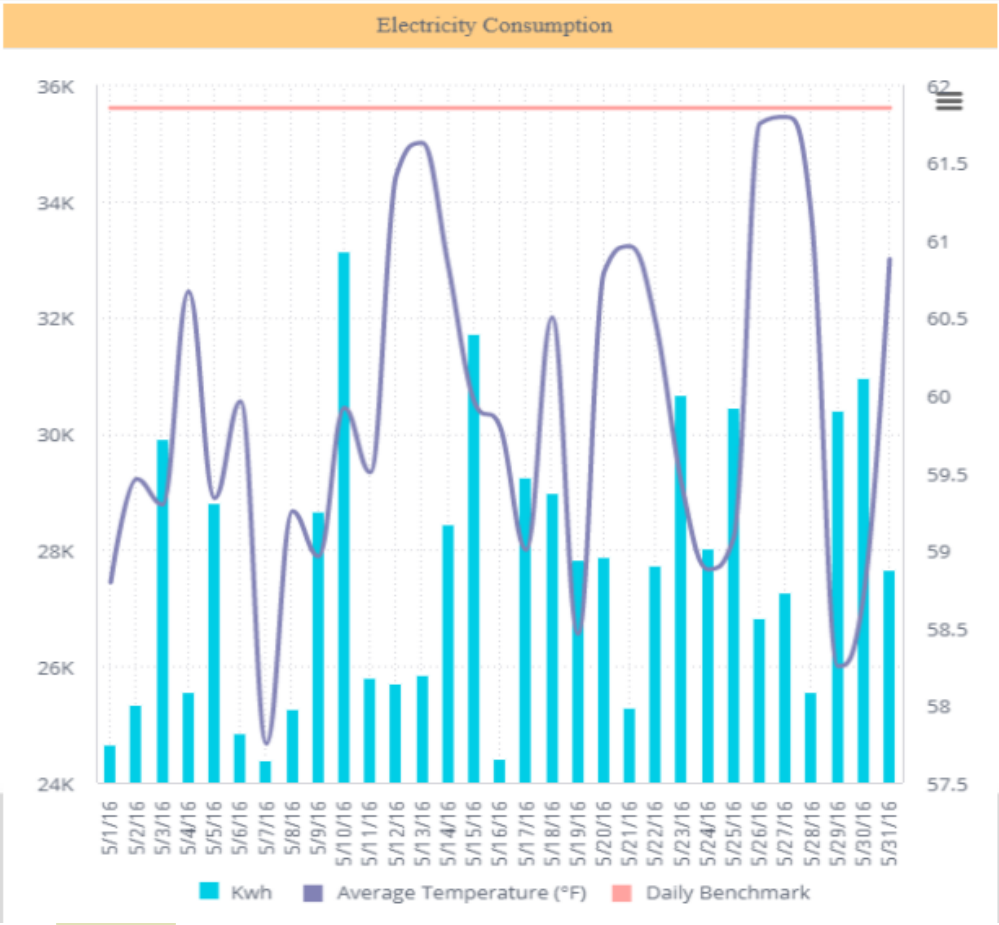
- "*Visual display of the most important information needed to achieve one or more objectives which fits entirely on a single computer screen so it can be monitored at a glance.*" (Few, 2013 p34)

- Small, concise, clear, intuitive and often interactive display.
- Customised for a specific person, group or function
- Focus on monitoring and measuring
- Consists of a combination of charts and tables

Report

- A document that presents information in an organised format for a specific audience and specific purpose
- Information is usually organised in a combination of graphic, tabular and / or narrative formats.
- Reports can be static or interactive (e.g. user can enter parameters, sort columns, filter content)

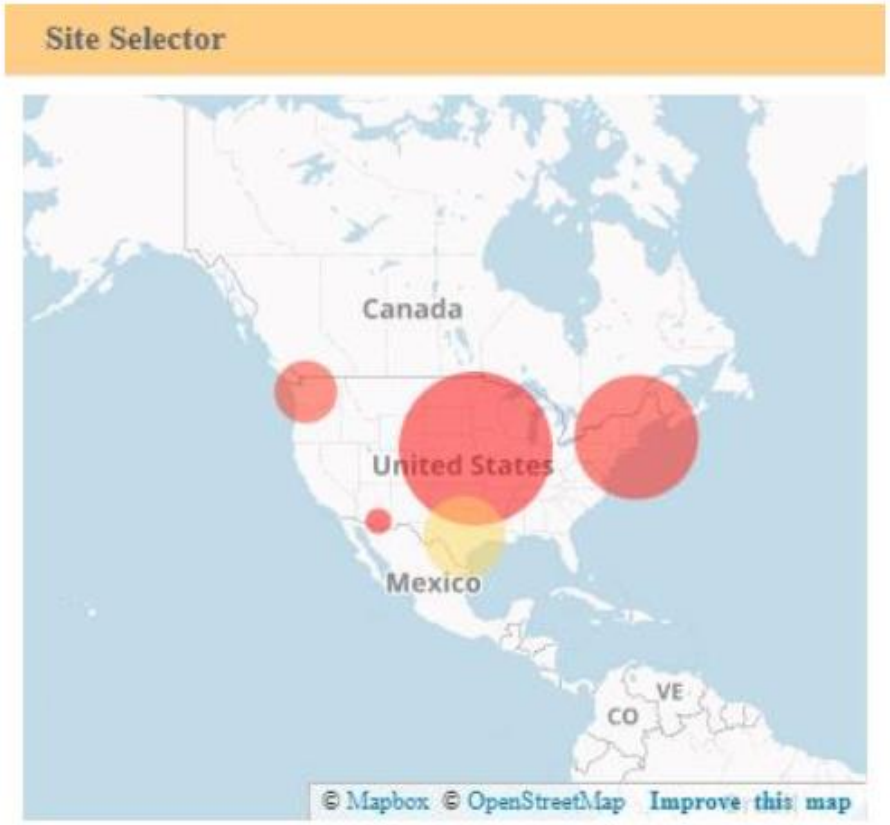
Report - Examples on Electricity Consumption



Days in Date	Kwh	Average Temperature (°F)	Daily Benchmark
05/1/2016	24657.708	58.79166667	35600
05/2/2016	25332.25	59.45833333	35600
05/3/2016	29905.583	59.29166667	35600
05/4/2016	25558.875	60.66666667	35600
05/5/2016	28812.958	59.33333333	35600
05/6/2016	24851.583	59.95833333	35600
05/7/2016	24391.25	57.75	35600
05/8/2016	25271.375	59.25	35600
05/9/2016	28669.917	58.95833333	35600
05/10/2016	33140	59.91666667	35600
05/11/2016	25810.625	59.5	35600
05/12/2016	25704.75	61.41666667	35600
5/13/16	25868.5	61.625	35600
5/14/16	28450.583	60.83333333	35600
5/15/16	31713.958	59.95833333	35600
5/16/16	24424.5	59.79166667	35600

Credit: sisense.com

Report - Examples on Electricity Consumption





Credit: sisense.com

Tables vs Graphs

- Tables are the most appropriate method to:
  - Display information when the user has the need to look up individual values.
  - Display simple relationship between numeric values and categorical values.

- Graphs are the most appropriate method when:
  - The message is contained in the shape of the values
  - The objective is used to reveal relationships among multiple values.

Types of Dashboards

Strategic	Analytic	Operational
• Used by managers to monitor business health and progress towards meeting strategic objectives • High-level measures of past and forecast performance	• Used by analysts as a way to gain access to a broad • range of more detailed information	• Used to monitor the operation of a factory or service • Display of information is real-time or near-real-time • Not only display measures but also displays warnings and alerts

Record Label Bench Marking

Taking into account the average of my revenue indicators, how does my record label compare to the industry standards (average)?

Overview Of My Record Label's Revenue Vs. the Overall Industry's Revenue

ADD TITLE

COLUMBIA

Current Year Revenue:

$26,737,850

Past Year Revenue:

$24,595,240

Industry

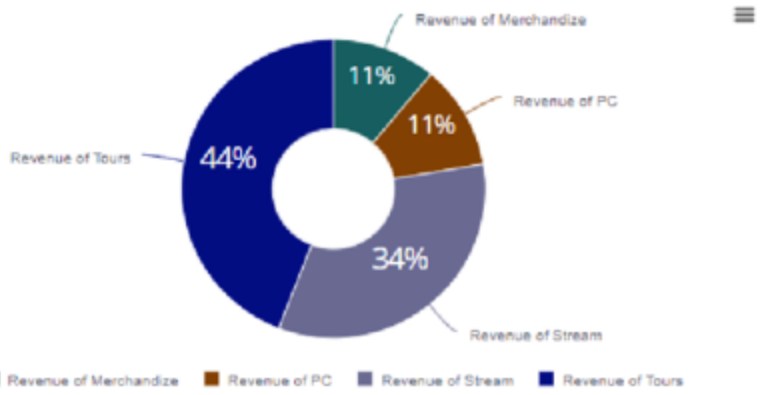
Current Year Revenue:

$129,582,257

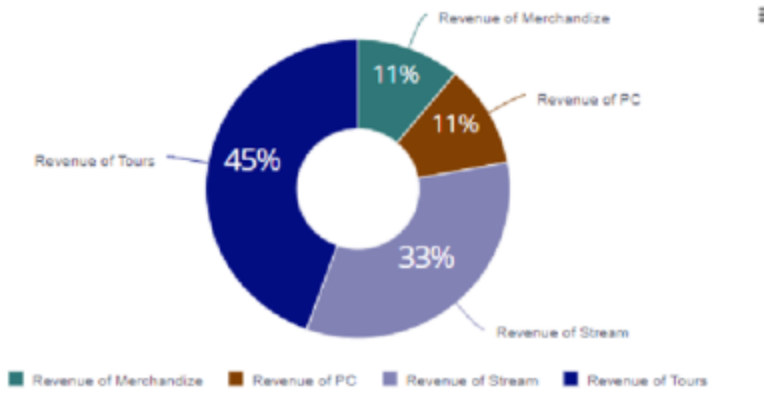
Past Year Revenue:

$145,047,224

Breakdown Of My Revenue Streams



Breakdown of the Industry's Revenue Streams



Student Attendance & Detention

2016-2017

As of 11/28/2016

ADD TITLE

Latest Enrollment

229

ADD TITLE

Attendance % YTD

90.26%

ADD TITLE

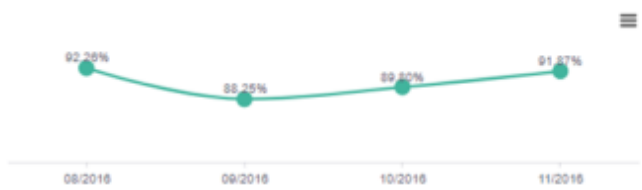
Suspension Rate YTD

3.49%

Enrollment by Month End



Attendance % By Month



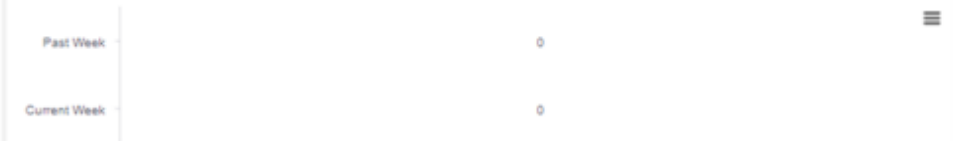
Suspension Rate By Month



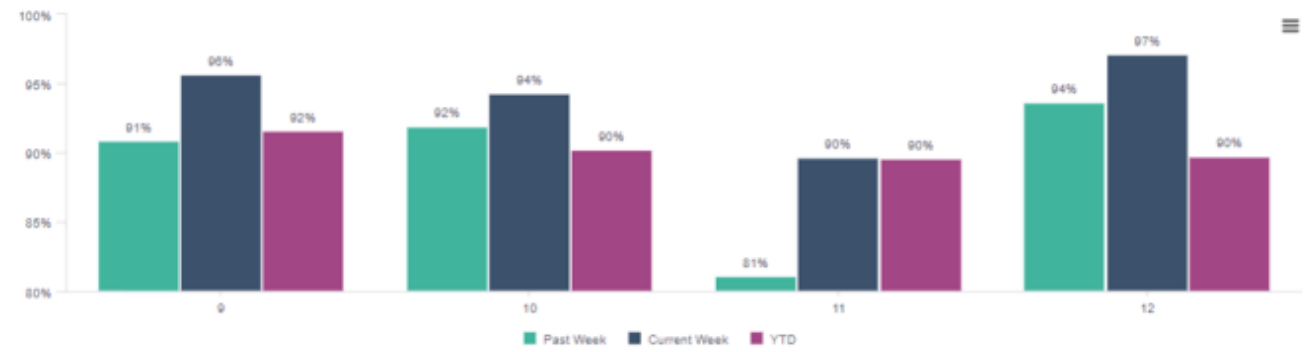
Attendance % - Current vs Past Week



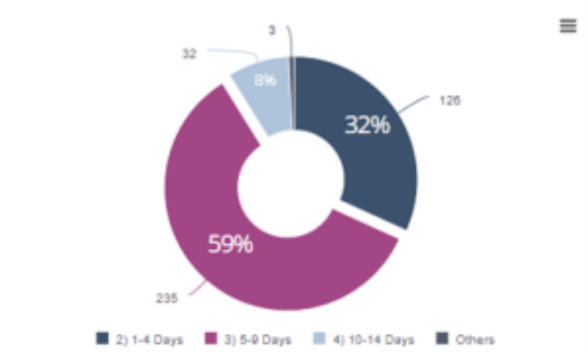
Number of Suspensions - Current vs Past Week



Attendance % By Grade



% Students Absent YTD



Dashboards versus Reports

- Minimal interactivity (Navigation should not be needed or at least kept to a minimum)
- Minimal analysis
- Require minimal cognitive effort
- Focus on quick communication of key data
- Individual visualisations should work together to communicate effectively.
- Click through to reports for more substantial analysis

BI Navigation Methods

- Drill-Down
  - Navigating to a report with more detail following the levels of a pre-defined dimension hierarchy.

- Example:

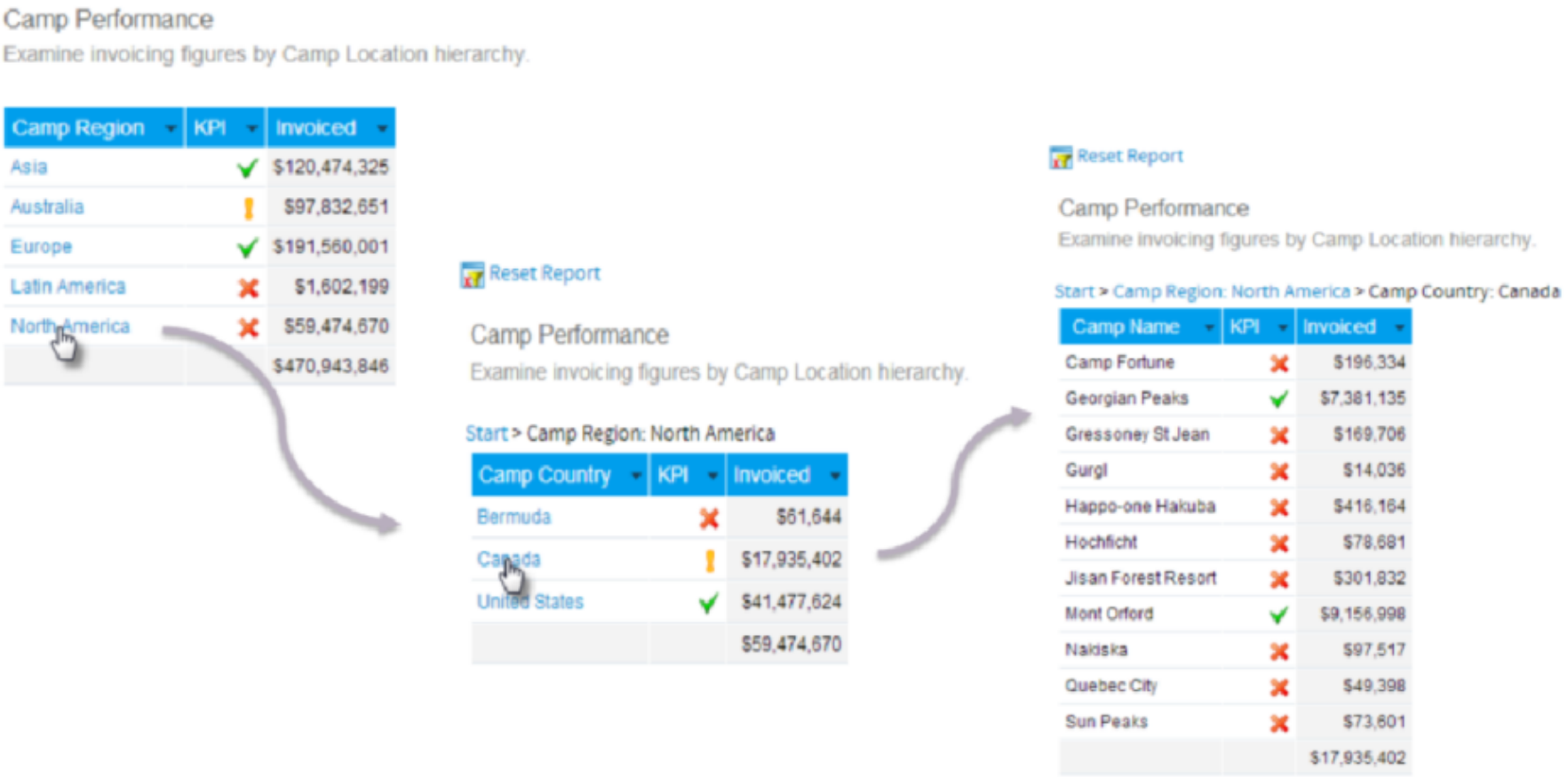
➔ Yearly sales

➔Quarterly sales

➔Monthly sales

➔Daily sales

Drill-Down - Example



BI Navigation Methods (cont.)

- Drill-Up (a.k.a. Roll-Up)
  - The reverse of “drill-down”
  - Navigating to a report with less detail following the levels of a pre-defined dimension hierarchy.

- Example:

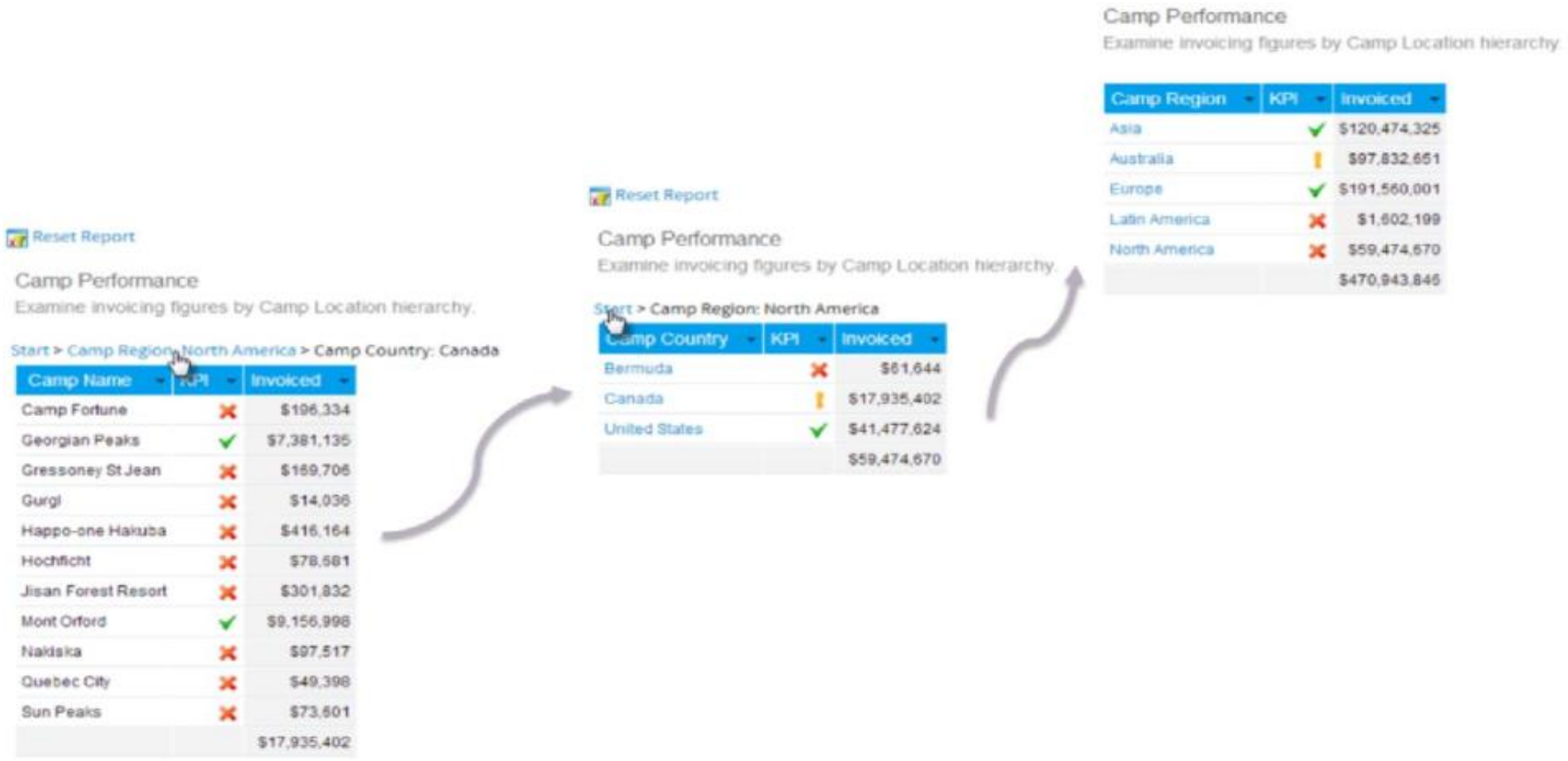
➔Yearly sales

➔Quarterly sales

➔Monthly sales

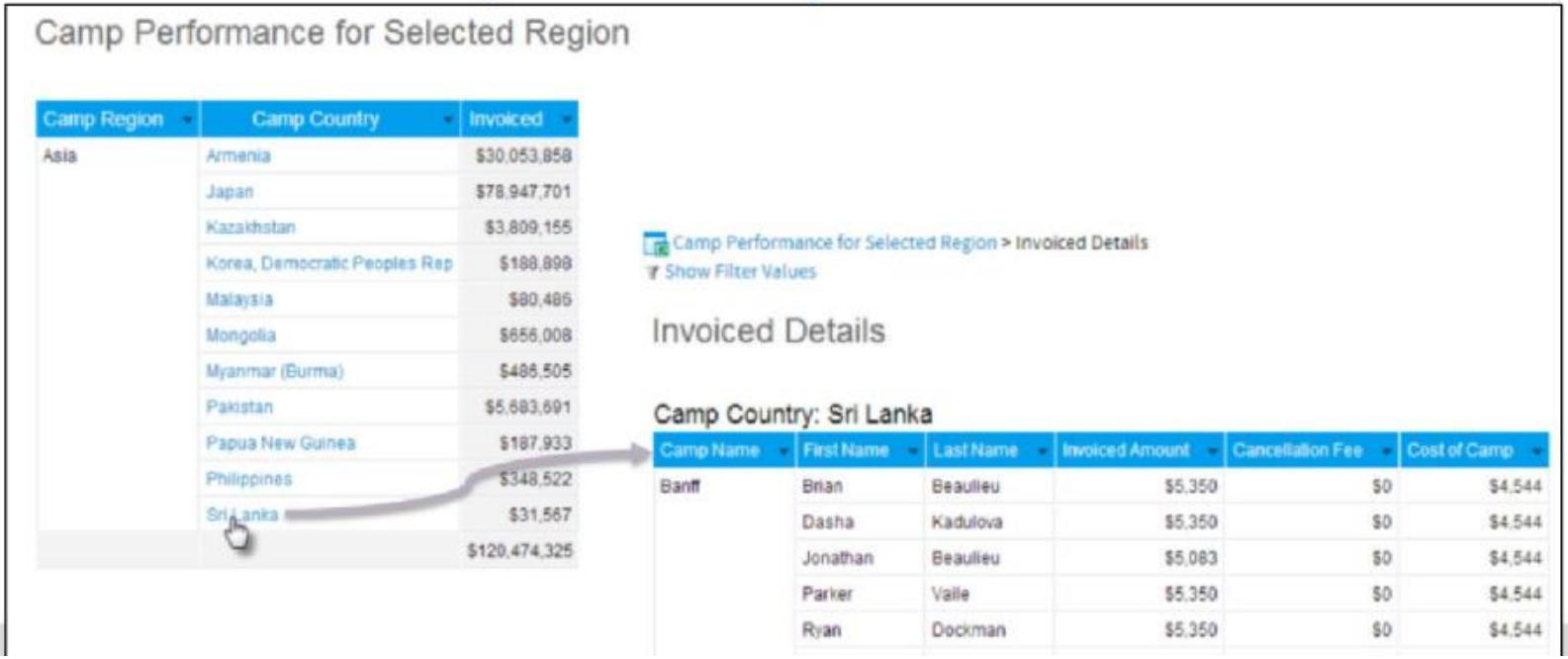
➔ Daily sales

Drill-Up - Example



BI Navigation Methods (cont.)

- Drill-Through
  - Navigating from a summary report to report that shows the detailed transactions that made up the summary.



BI Navigation Methods (cont.)

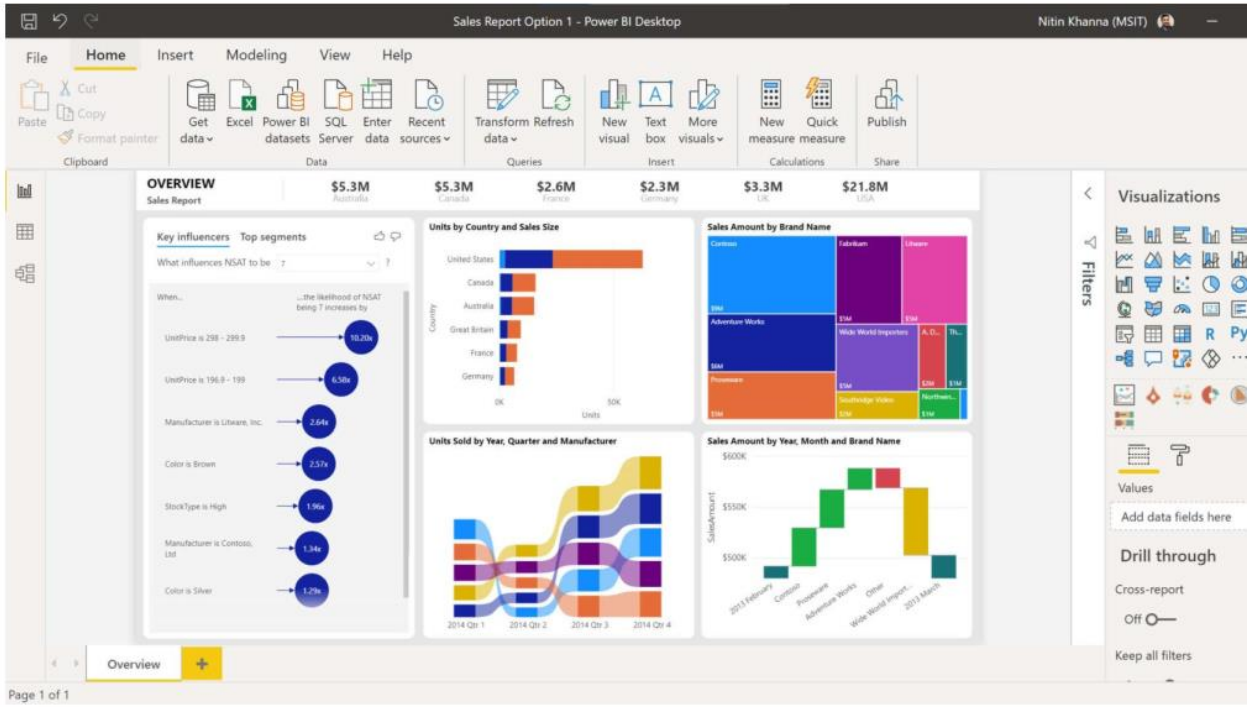
- Drill-Across
  - Navigating from a report based on one set of dimension attributes and measure(s) to a new report with the same set of dimension attributes but with different measure(s).



Power BI

Power BI is a Microsoft's business intelligence tool that comprises both desktop and web-based applications for loading, modeling, and visualizing data.[1]

Connect to and visualise any data using the unified, scalable platform for self-service and enterprise business intelligence (BI) that's easy to use and helps gain deeper data insight.[2]



Power BI

Bridge the gap between data and decision making [2]

1. Create amazing data experiences

   Easily connect to, model, and visualise your data, creating memorable reports personalised with your KPIs and brand. Get fast, AI-powered answers to your business questions—even when asking with conversational language.

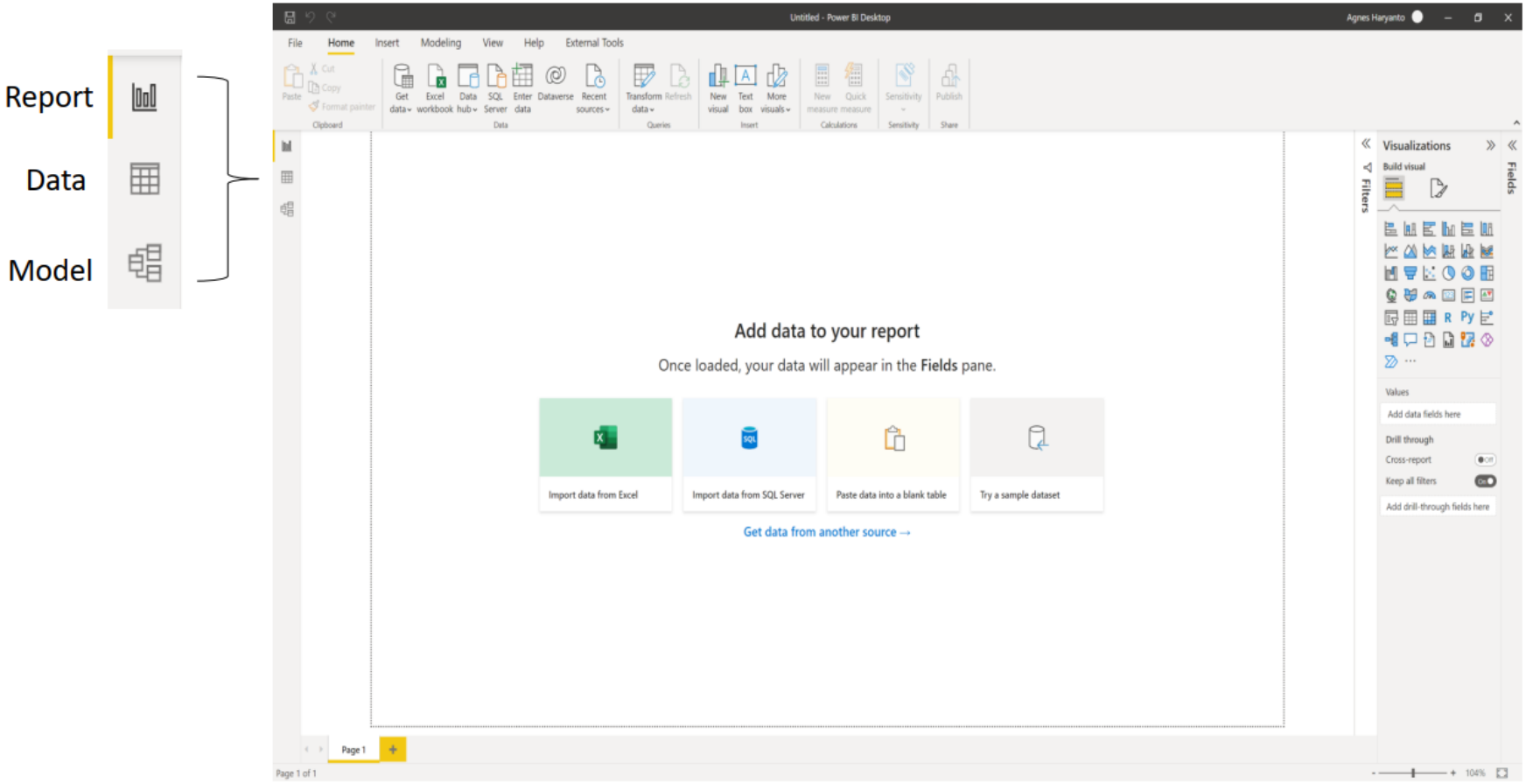
2. Gain insight from your largest BI deployments

   Make the most of your big data investments by connecting to all your data sources with the scale to analyse, share, and promote insights across your organisation while maintaining data accuracy, consistency, and security.

3. Make decisions with confidence

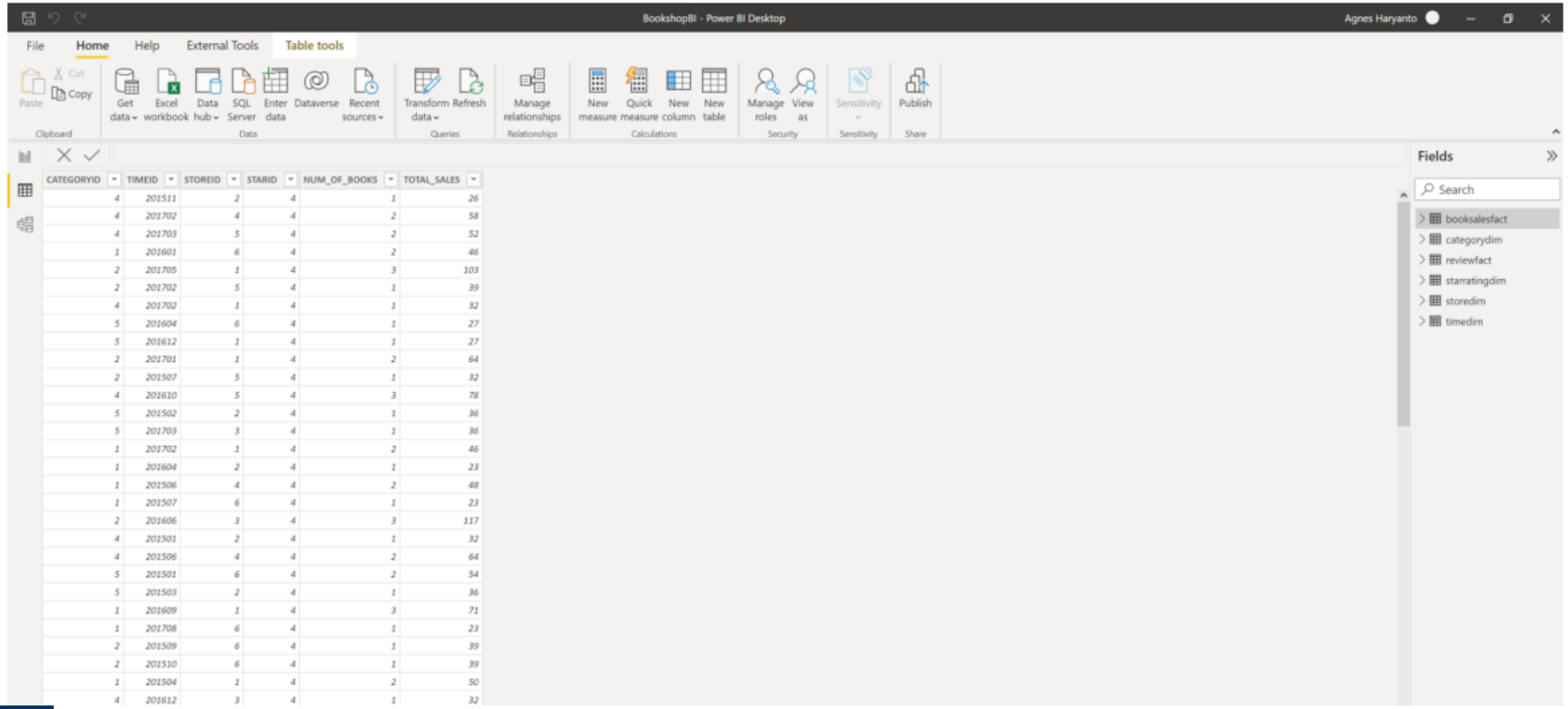
   Work together easily on the same data, collaborate on reports, and share insights across popular Microsoft Office applications such as Microsoft Teams and Excel—empowering everyone in your organisation to quickly make data-driven decisions that drive strategic actions.

Power BI Interface



Power BI Interface

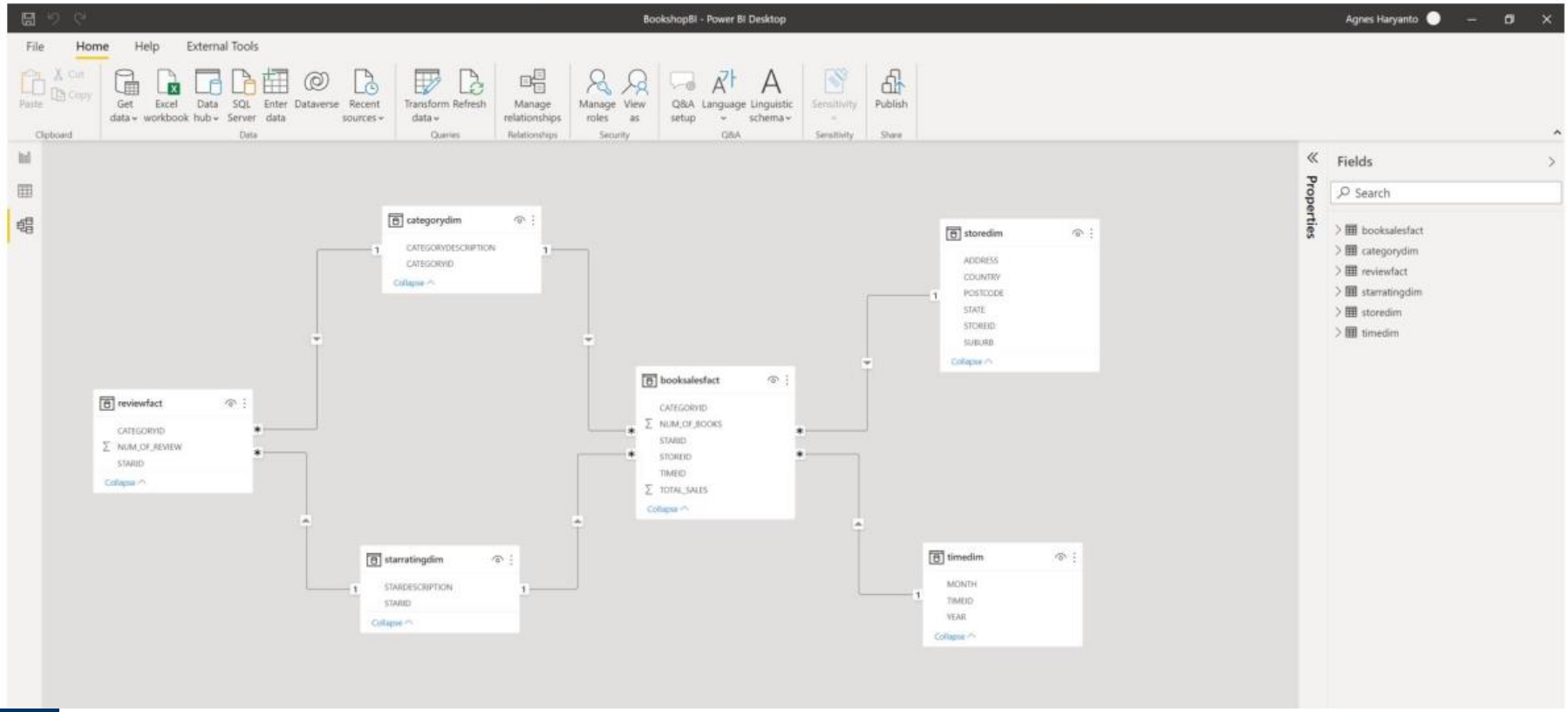
Data - *Connect, shape and transform raw data*



CATEGORYID	TIMEID	STOREID	STARID	NUM_OF_BOOKS	TOTAL_SALES
4	201511	2	4	1	26
4	201702	4	4	2	58
4	201703	5	4	2	52
1	201601	6	4	2	46
2	201705	1	4	3	103
2	201702	5	4	1	39
4	201702	1	4	1	32
5	201604	6	4	1	27
5	201612	1	4	1	27
2	201701	1	4	2	64
2	201507	5	4	1	32
4	201610	5	4	3	78
5	201502	2	4	1	36
5	201703	3	4	1	36
1	201702	1	4	2	46
1	201604	2	4	1	23
1	201506	4	4	2	48
1	201507	6	4	1	23
2	201606	3	4	3	117
4	201501	2	4	1	32
4	201506	4	4	2	64
5	201501	6	4	2	54
5	201503	2	4	1	36
1	201609	1	4	3	71
1	201708	6	4	1	23
2	201509	6	4	1	39
2	201510	6	4	1	39
1	201504	1	4	2	50
4	201612	3	4	1	32

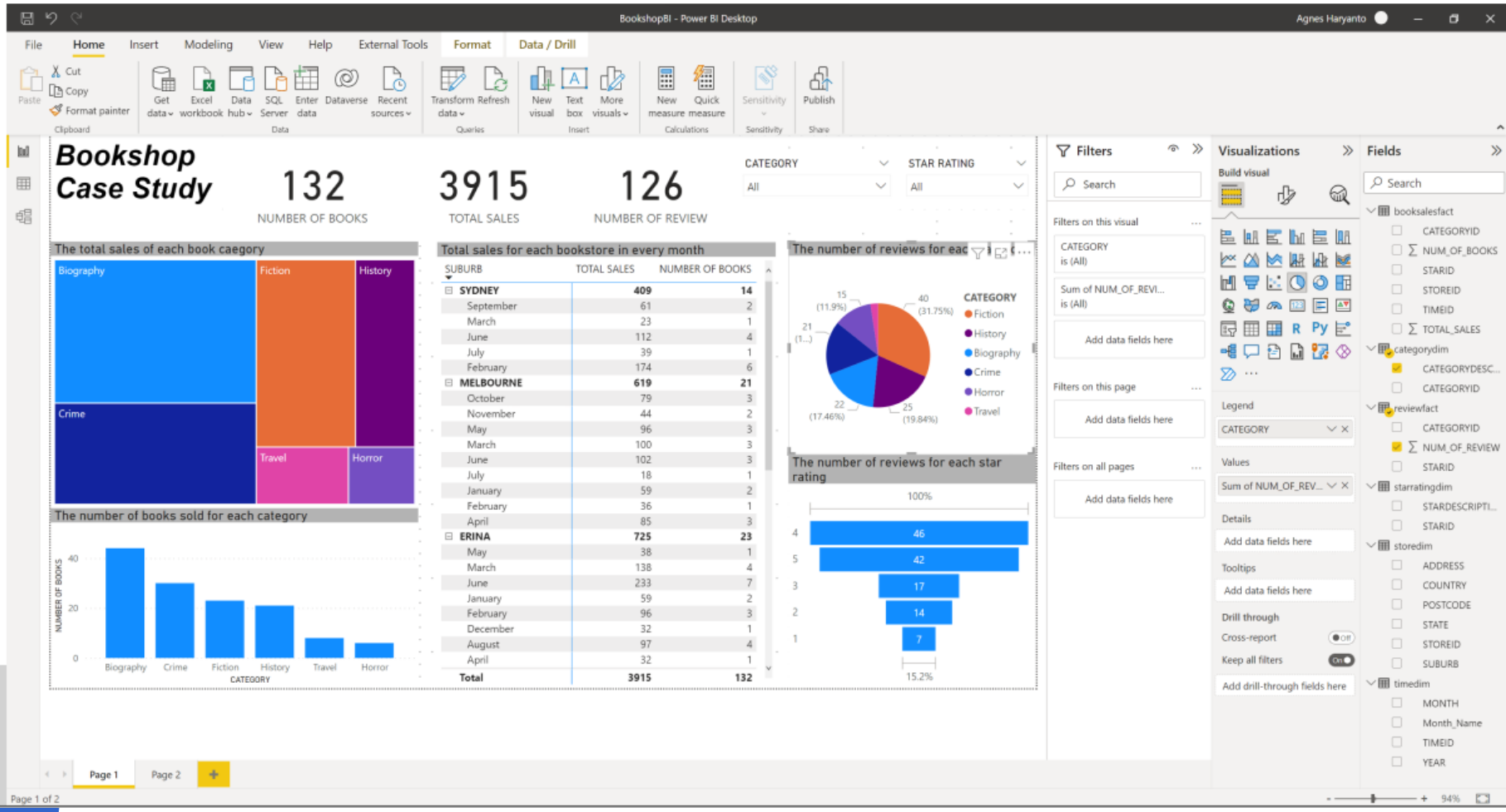
Power BI Interface

Model - *Build table relationships to create a data model design*



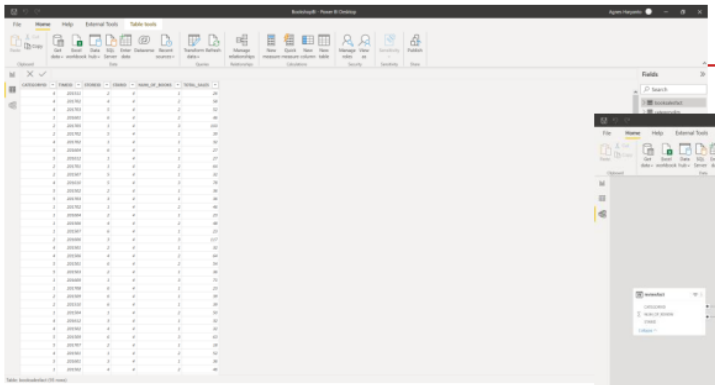
Power BI Interface

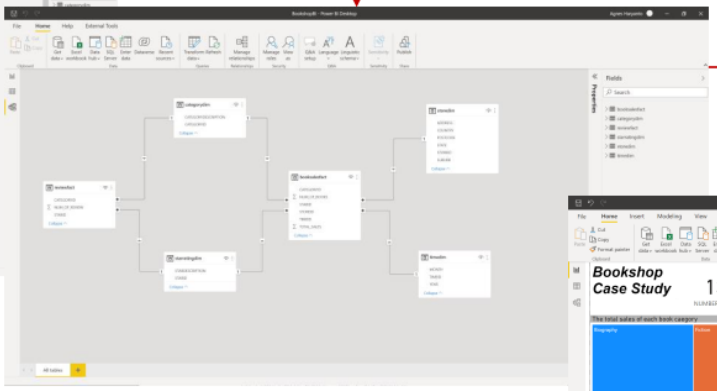
Report - *Design interactive reports to explore and visualize data*

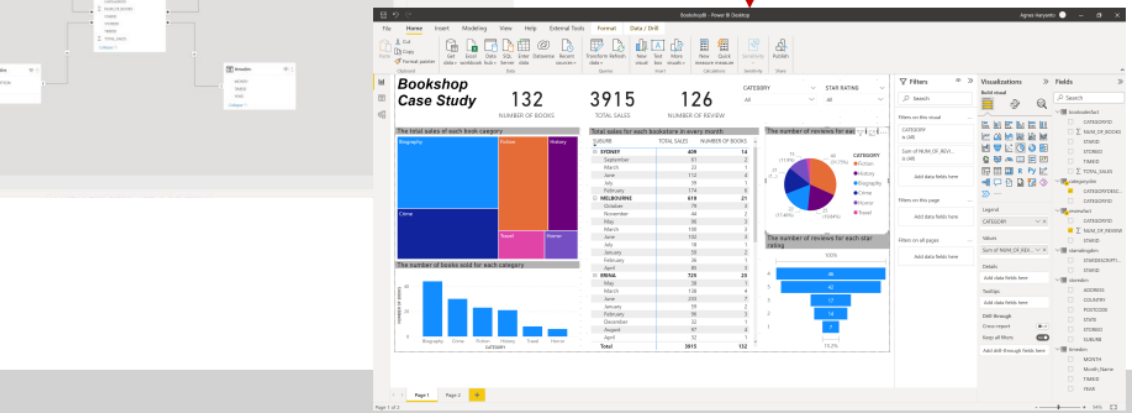


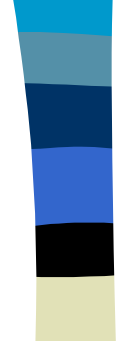
Power BI Workflow

Data → Model → Report









The Book Sales Case Study

The system stores information about books, including the authors, publishers, book categories, as well as the reviews that each book has received. The ”stars” attribute in the Review entity records the star rating for each review (e.g. 5 stars for excellent to 1 star for poor, etc). One book may receive many reviews. For simplicity, it is assumed, as also shown in the E/R diagram, that a book will only have one category.

The E/R diagram also includes entities related to sales of books, and the stores which sale the books. Each store has many sales transactions (i.e. the Sales entity), and each sales transaction may include several books (i.e. the SalesDetails entity). The Total Price attribute in the Sales Details entity is basically Quantity multiplied by the Unit Price, whereas Total Price in the Sales entity is the total price for each sales transaction.





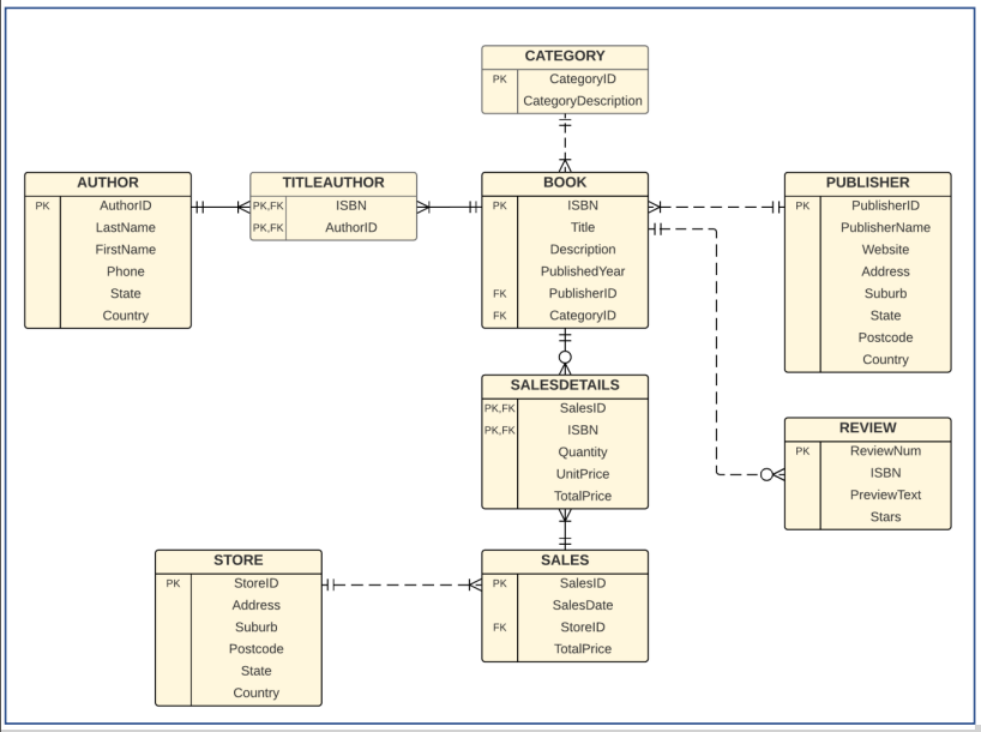
The Book Sales Case Study

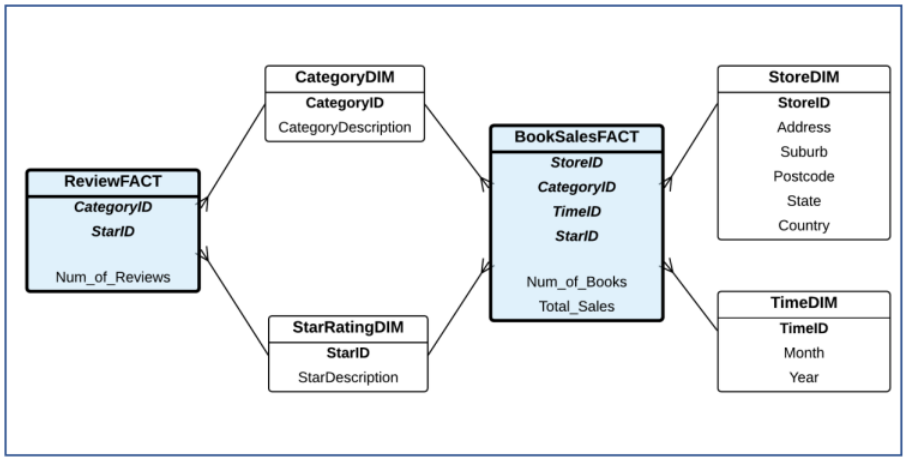
The requirements for the data warehouse are quite simple. The data warehouse must be able to answer at least the following questions:

- What are the total sales for each bookstore in a month?
- What is the number of books sold for each category?
- What is the book category that has the highest total sales?
- What is the number of reviews for each category?
- How many 5-star reviews for each category?

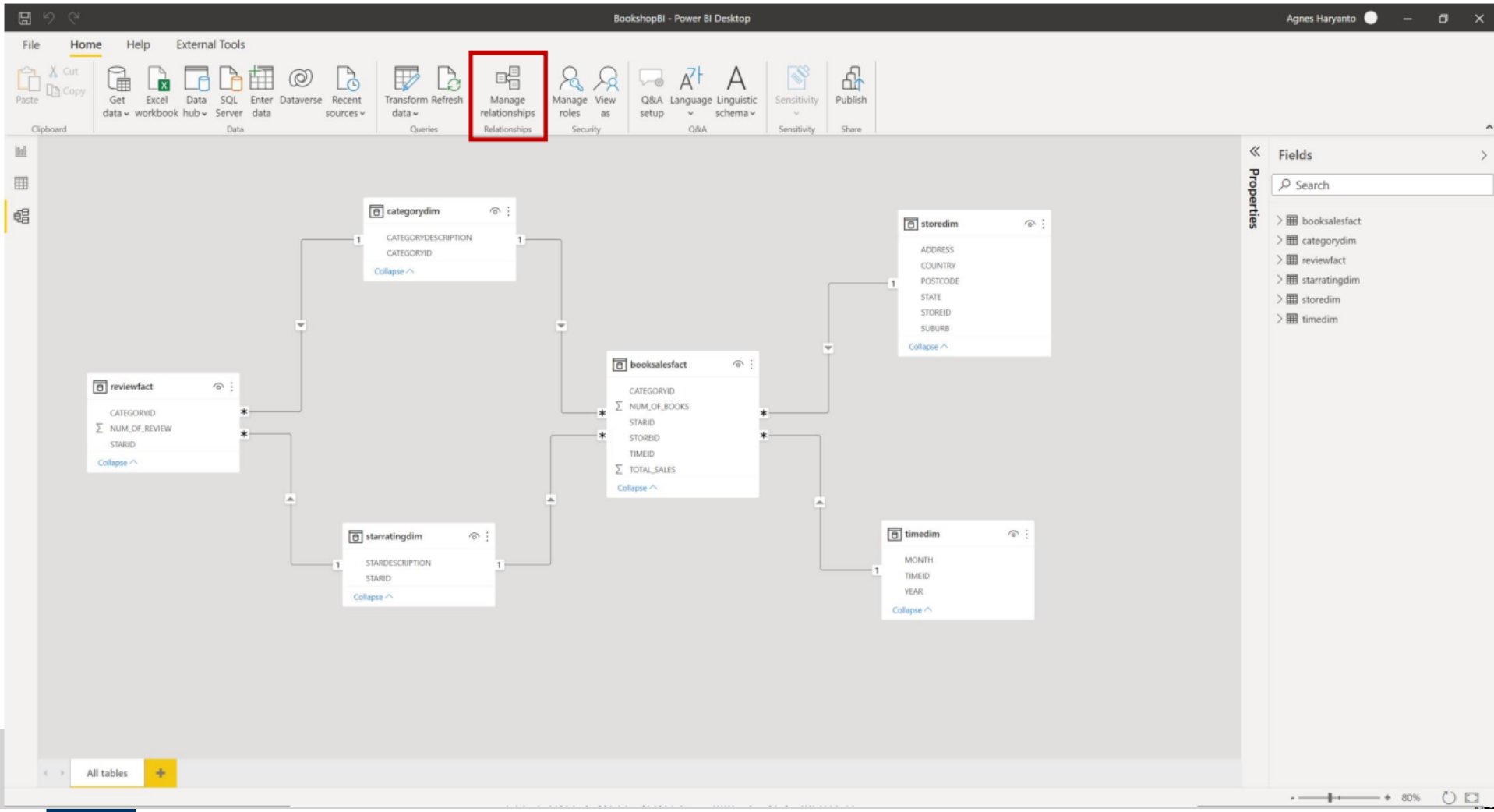


The Book Sales Case Study

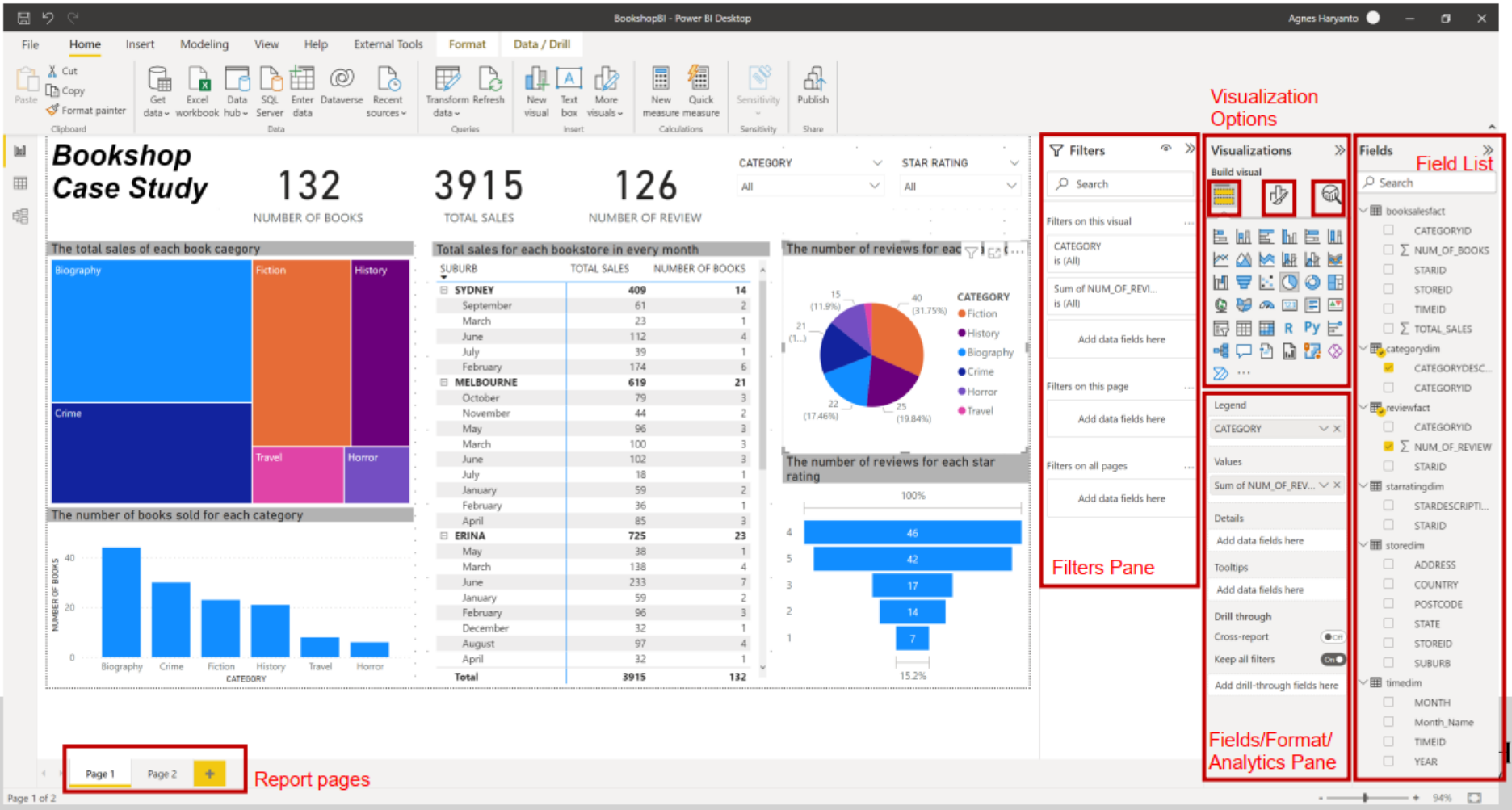




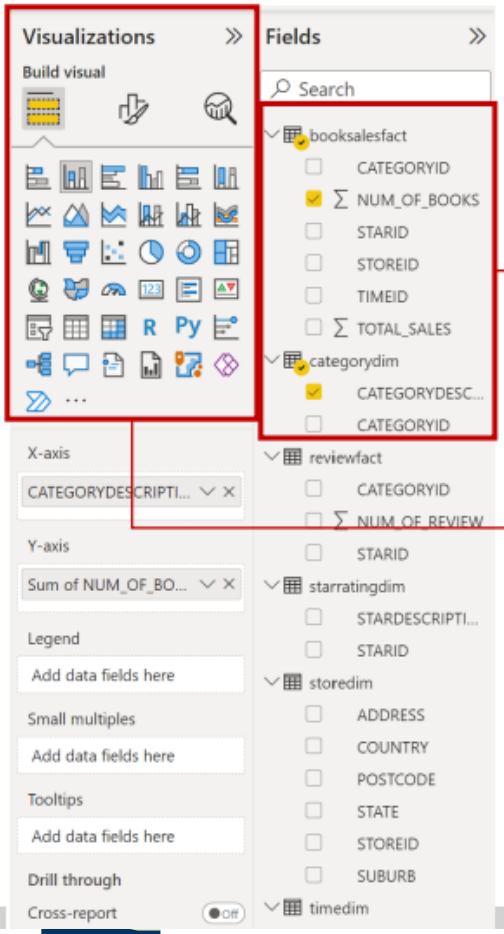
Model View Tab

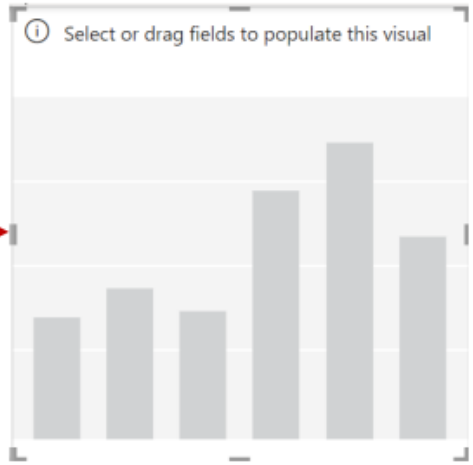


Power BI Report View

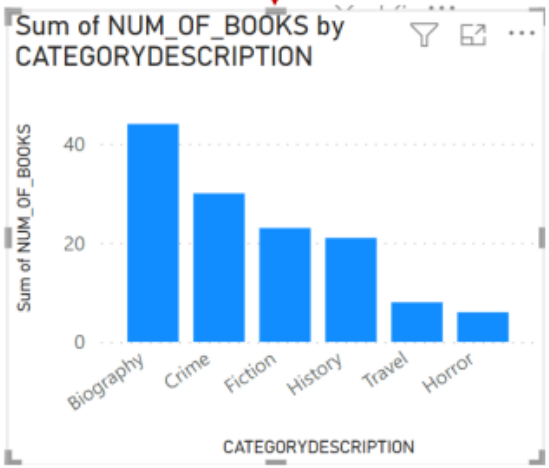


Start visualizing data



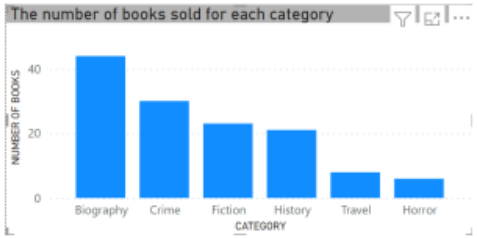


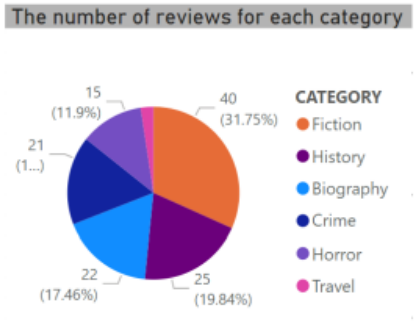
Choose the visualization type to start a blank chart template

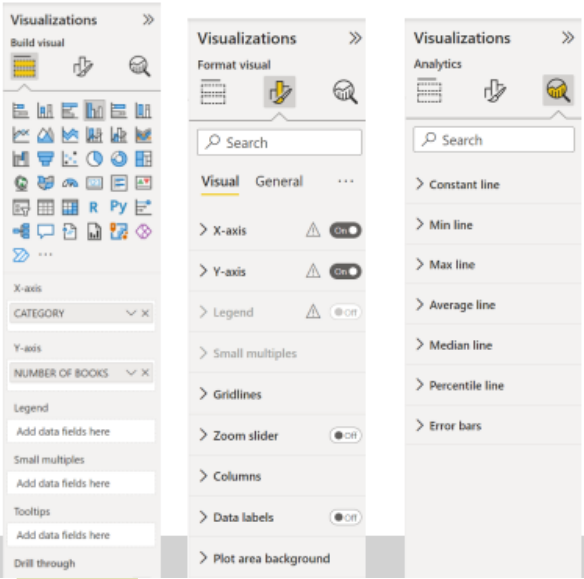


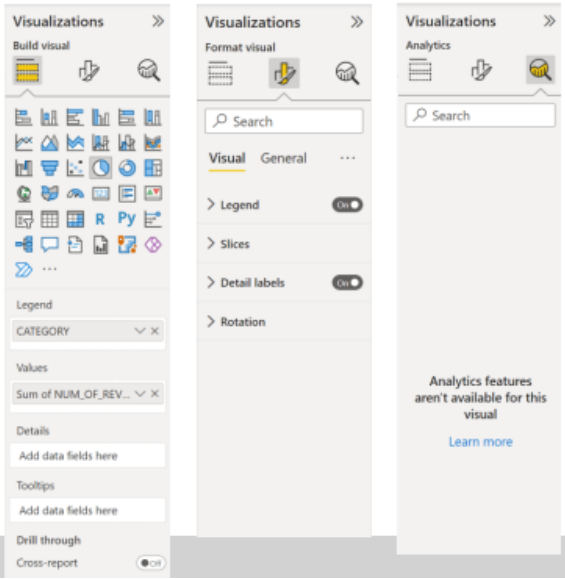
Choose attributes into the report canvas to generate a new visual

Formatting Options









Note: Each visualization would have different formatting options

The Book Sales Dashboard

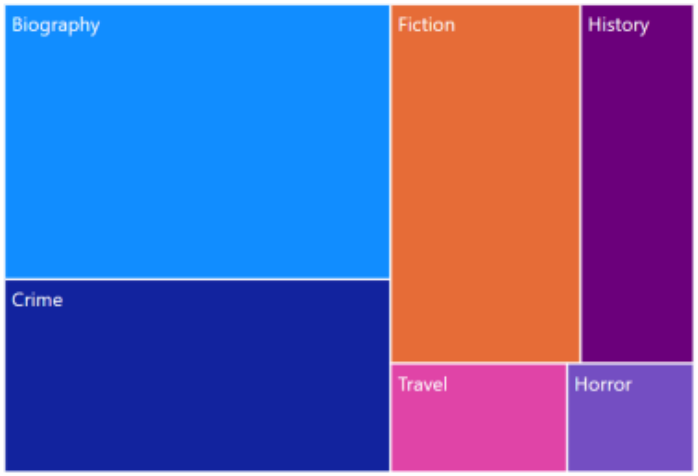
Bookshop Case Study

132	3915	126
NUMBER OF BOOKS	TOTAL SALES	NUMBER OF REVIEW

CATEGORY: All

STAR RATING: All

The total sales of each book caegory



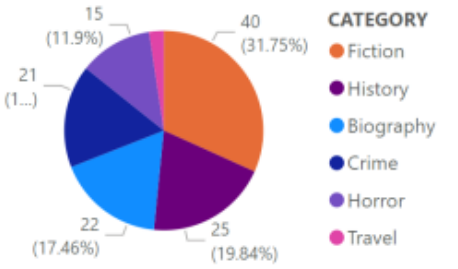
The number of books sold for each category



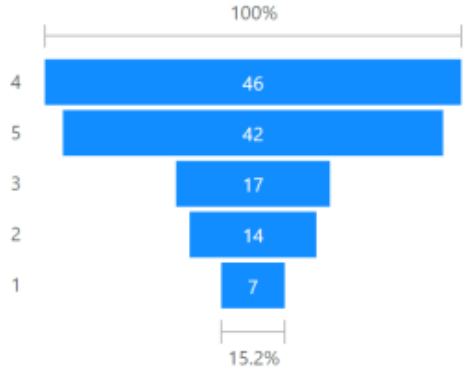
Total sales for each bookstore in every month

SUBURB	TOTAL SALES	NUMBER OF BOOKS
SYDNEY	**409**	**14**
September	61	2
March	23	1
June	112	4
July	39	1
February	174	6
MELBOURNE	**619**	**21**
October	79	3
November	44	2
May	96	3
March	100	3
June	102	3
July	18	1
January	59	2
February	36	1
April	85	3
ERINA	**725**	**23**
May	38	1
March	138	4
June	233	7
January	59	2
February	96	3
December	32	1
August	97	4
April	32	1
Total	**3915**	**132**

The number of reviews for each category



The number of reviews for each star rating



Desktop vs Mobile Layouts

